Reproductive System

Anatomical and Clinical Foundation

The reproductive system is a specialized complex responsible for the **production of gametes** (spermatozoa and oocytes), the **secretion of sex hormones**, **fertilization**, and in females, the **support of embryonic and fetal development**. Structurally, it is organized into primary sex organs (gonads), genital ducts, accessory glands, and external genitalia.

I. Male Reproductive System

The male system consists of the testes, a series of excretory ducts, accessory glands, and external genitalia.

1. The Testes (Gonads)

- **Location:** Situated in the **scrotum**, outside the abdominopelvic cavity.
- **Functions:** Dual role involving **spermatogenesis** (sperm production) and the synthesis of **androgen hormones**.
- **Morphology:** Oval-shaped organs, approximately 4–5 cm in length and weighing 15–25 g. They possess superior and inferior poles, and anterior and posterior borders. The **posterior border** is specifically related to the epididymis.
- **Internal Structure:** The **tunica albuginea** is a fibrous capsule that forms septa, dividing the testis into lobules. Each lobule contains 1–4 **seminiferous tubules** (site of sperm production) and **interstitial tissue** containing **Leydig cells** (responsible for hormone production).

2. Excretory Duct System

- **Epididymis:** Located on the posterior aspect of the testis. It is divided into three parts and serves as the site for **sperm maturation** and **storage**.
- **Vas Deferens (Ductus Deferens):** A muscular tube approximately **45 cm** long. It begins at the tail of the epididymis, ascends within the **spermatic cord**, passes through the **inguinal canal**, and enters the pelvis. It eventually joins the duct of the seminal vesicle to form the **ejaculatory duct**.

3. Accessory Glands

- **Seminal Vesicles:** Paired glands (approx. 5 cm) located posterior to the bladder. They secrete an **alkaline, fructose-rich fluid** that constitutes a significant portion of semen.
- **Prostate Gland:** A walnut-sized organ located in the **lesser pelvis**, inferior to the urinary bladder and anterior to the rectum. It surrounds the **proximal urethra** and produces alkaline fluid. Zones: Central, Fibromuscular, Transitional, Peripheral, and Periurethral.

4. External Genitalia: The Penis

External Structure: Comprises the **Shaft** (body), the **Glans** (sensitive tip), the **Prepuce** (foreskin), and the **Urinary Meatus** (urethral opening at the tip).

Internal Structure (Erectile Tissues):

- **Corpora Cavernosa:** Two paired cylinders on the dorsal (upper) side.
- **Corpus Spongiosum:** A single cylinder on the ventral (lower) side.
- **Tunica Albuginea:** A tough membrane wrapping around the corpora cavernosa.

II. Female Reproductive System

The female system includes the ovaries, uterine tubes, uterus, vagina, and vulva.

1. The Ovaries (Gonads)

- **Characteristics:** Paired, almond-shaped glands measuring approximately **3 × 1.5 × 1 cm**.
- **Location:** Situated in the pelvic cavity, lateral to the uterus.
- **Function:** Responsible for producing **oocytes** and hormones (**estrogen** and **progesterone**).

2. Uterine Tubes (Fallopian Tubes)

Paired tubes, 10–12 cm long, that convey oocytes from the ovaries to the uterus.

- **Fimbriae:** Finger-like projections at the end.
- **Infundibulum:** The funnel-shaped opening.
- **Ampulla:** The widest part and the **usual site of fertilization**.
- **Isthmus:** The narrow connection to the uterus.

3. The Uterus

- **Structure:** A hollow, muscular, pear-shaped organ (7–8 cm long, 5 cm wide) located between the **bladder** (anterior) and the **rectum** (posterior).
- **Divisions: Fundus** (top), **body** (main part), **isthmus** (constriction), and **cervix** (neck).

4. Vagina and Vulva

- **Vagina:** A fibromuscular canal (8–10 cm). Located anterior to the rectum and posterior to the bladder and urethra.
- **Vulva (External Genitalia):** Provides protective covering for reproductive and urinary openings. Includes the **mons pubis**, **clitoris**, **inner and outer labia**, and the **vaginal and urethral openings**.

III. Anatomical Basis for Pharmacy

Anatomical knowledge is essential for:

- Predicting **drug distribution** and understanding **hormonal drug actions**.
- Recognizing **cancer spread patterns** and advising on **contraceptive methods**.
- Utilizing **local drug delivery** routes, such as the vaginal route.

MCQ Notes

Male System Traps

- **Penis Anatomy:** The penis contains **two** corpora cavernosa and **one** corpus spongiosum; exam questions often flip these numbers.
- **Ductal Formation:** The **ejaculatory duct** is formed by the fusion of the vas deferens and the seminal vesicle duct. It is a mistake to say the vas deferens is formed by this fusion.
- **Prostate Passage:** Only the **urethra** and **ejaculatory ducts** traverse the prostate. The **ureters** (connecting kidneys to bladder) do **not** enter the prostate.
- **Testes:** They are **mixed glands** (exocrine for sperm, endocrine for testosterone). They are **external** organs (scrotum), and they do **not** secrete estrogens.

Female System Traps

- **Uterus Location:** It is in the **median** (central) position of the pelvis. It is **not** anterior or superior.
- **Fertilization vs. Gestation:** Fertilization occurs in the **uterine tubes** (Fallopian tubes), **not** in the uterus or ovaries. The uterus is the site of **gestation** (nidation).
- **Urethral Meatus:** In females, the urethral opening is located **anterior** to the vaginal opening, situated between the clitoris (above) and the vagina (below).
- **Ovaries:** Attached to the uterus by the **tubo-ovarian ligament**. Almond-shaped (not kidney/bean-shaped) and contain a **finite, non-renewable** stock of oocytes.
- **Internal vs. External:** The **vulva** is an external organ; the uterus, tubes, vagina, and ovaries are internal.
- **Breast Anatomy:** The **areola** is the pigmented skin area that **surrounds** the mamelon (nipple); it is not the other way around.